

TARUN KUMAR RAGHAV

DATA SCIENTIST | MACHINE LEARNING | AI / GENAI | DATA ENGINEERING

Thane, India | raghavtarunkumar@gmail.com | +91-8286572865 | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [HackerRank](#)

PROFESSIONAL SUMMARY

Data professional with 3+ years in banking, fintech and loyalty-rewards analytics, combining a strong data engineering base (Python, SQL, PySpark, ETL) with applied data science: customer feature engineering, EDA, RFM segmentation and scikit-learn regression/classification. Prepares ML-ready datasets from transaction, points and redemption data for segmentation, predictive-modelling and cross-selling use cases. Built an NLP resume-screening API (TF-IDF, FastAPI) and a RAG financial-document Q&A app (LangChain, FAISS). Promoted to Senior Software Engineer for end-to-end pipeline ownership.

TECHNICAL SKILLS

Programming: Python, SQL

Data Science & ML: Statistics, EDA, Feature Engineering, Regression, Classification, Clustering, Model Evaluation, Cross-Validation, Predictive Modelling, RFM Segmentation, Customer Analytics

NLP & GenAI: Prompt Engineering, RAG, LangChain, FAISS, Embeddings, Semantic Search, LLM APIs, TF-IDF, Cosine Similarity

Python Libraries: Pandas, NumPy, Scikit-learn, PySpark, Spark SQL, Matplotlib, Seaborn

Data Engineering: ETL/ELT, Data Pipelines, Data Validation, Data Transformation, Data Modelling, SFTP, SSIS, Azure Data Factory

Databases: PostgreSQL, MySQL, SQL Server, MongoDB

BI & Analytics: Power BI, DAX, SSRS, Advanced Excel, KPI Development, Statistical Analysis

APIs & Tools: FastAPI, Flask, SQLAlchemy, REST APIs, Streamlit, Git, GitHub, Postman, Google Colab

Domain: Banking, Fintech, Loyalty & Rewards, Transaction Analytics, Points & Redemption

PROFESSIONAL EXPERIENCE

Loylty Rewardz

Thane, India

Senior Software Engineer

Apr 2026 – Present

- Promoted from Software Engineer (Apr 2026) for consistent delivery of scalable data pipelines and automation initiatives supporting banking and loyalty analytics.
- Engineer customer-level behavioural features from transaction, partner, category, MCC, points, redemption and geographic data to support customer segmentation, predictive modelling and cross-selling use cases.
- Prepare ML-ready datasets and apply EDA and regression/classification fundamentals to customer-behaviour and redemption-opportunity analysis, supporting predictive-modelling and recommendation use cases.
- Lead design and development of Python, SQL and PySpark solutions that consolidate multi-bank transaction, customer, account, points, redemption and store data into a Customer 360 foundation for analytics and ML workflows.
- Own data pipeline architecture from SFTP ingestion through validation, transformation and data modelling to a KPI layer used for reporting and business decision support.
- Drive data-quality practices — validation rules, reconciliation checks, merchant/partner/MCC standardization and live-data null handling — so downstream features and KPIs stay reliable.
- Mentor engineers on ETL design, data validation and reporting automation, and partner with business stakeholders to translate loyalty, redemption and partner-engagement goals into data solutions.

Software Engineer

Mar 2025 – Apr 2026

- Extracted, transformed and structured large-scale banking datasets with SQL (MySQL, PostgreSQL), enabling feature engineering and analytical dataset creation for machine learning workflows.
- Built scalable Python (Pandas, NumPy) and PySpark workflows for large transactional data, improving the efficiency of data preparation for analysis and modelling.
- Developed SSIS and SFTP pipelines for reliable ingestion of high-quality data used for EDA, reporting and ML use cases.
- Standardized and validated transaction and customer data: regex-based city normalization (e.g., Mumbai / Bombay / Mum), city-to-state mapping, business-rule geographic imputation, and merchant, partner, corporate and MCC standardization.
- Performed EDA in Python and Excel to identify trends, anomalies and behavioural patterns in customer and transaction data, supporting data-driven business decisions.
- Applied regression/classification concepts to business problems and prepared datasets for predictive-modelling use cases such as customer-behaviour analysis.
- Designed Power BI and Excel dashboards visualizing KPIs and insights for stakeholders, enabling data-driven strategy decisions.
- Conducted RCA and data validation with SQL and Python to improve data quality and downstream accuracy; built Flask/SQLAlchemy automation tools and APIs that reduced manual data-processing effort.

Tech Stack: Python (Pandas, NumPy), SQL (MySQL, PostgreSQL), PySpark, SSIS, SFTP, Flask, SQLAlchemy, MongoDB, Power BI, Excel

Creative Information Technology, Inc. (CITI)

Data Analyst

Jun 2023 – Mar 2025

- Executed data cleaning and transformation in Azure Data Factory, improving data accuracy by **30%** for business reporting.
- Developed Python (Pandas) scripts for file processing, cutting data processing time by **40%** and boosting workflow efficiency.
- Wrote and optimized daily MSSQL queries for reporting, improving management decision-making speed by **25%**.
- Conducted data validation and reporting that achieved a **20%** reduction in data-related issues reported by business users; used advanced Excel for data cleaning and presentation, reducing manual reporting effort by **35%**.
- Designed interactive Power BI dashboards for **50+** business users and resolved **1,000+** dashboard and data-request support tickets in close collaboration with stakeholders.

Tech Stack: MSSQL, Azure Data Factory, Python (Pandas), Power BI, Excel, Postman

SELECTED DATA SCIENCE / ML / AI PROJECTS

FinDoc AI — GenAI Financial Document Intelligence System | Python, Streamlit, LangChain, FAISS, LLM APIs

- Built a Retrieval-Augmented Generation (RAG) application enabling natural-language question answering over financial and business documents.
- Developed the end-to-end pipeline: PDF extraction, text chunking, embedding generation, FAISS vector indexing, semantic retrieval and LLM-based response generation.
- Implemented source-aware answers with document/page references, grounding responses in retrieved context to reduce hallucinations; added conversational querying and document summarization.

AI Resume Screening System (Personal Project) | Python, NLP, Scikit-learn, FastAPI

- Built an NLP-based resume-screening tool using TF-IDF vectorization and cosine similarity to generate ATS-style candidate match scores.
- Exposed FastAPI endpoints for resume upload, parsing and scoring, delivering the tool as an end-to-end API service.

Bank KPI & Analytics Dashboard | PostgreSQL, SQL, Power BI, DAX, Excel

- Delivered an end-to-end customer analytics solution by integrating cleaned month-on-month (MoM) banking data into PostgreSQL, with SQL transformations, data modelling and DAX measures feeding interactive Power BI dashboards.
- Built KPI, RFM customer segmentation, cross-selling and advanced statistical dashboards to uncover customer behaviour, spending patterns and revenue-growth opportunities, with drill-through views for executive monitoring and trend analysis.

Partner Health WoW & MoM Reporting Automation | Python, PostgreSQL, SSRS, SMTP

- Built an end-to-end automated reporting pipeline (PostgreSQL dynamic SQL, SSRS, Python with PyMuPDF and SMTP) that prepares week-on-week (WoW) and month-on-month (MoM) partner health metrics and emails dashboard-style reports to stakeholders, improving reporting efficiency and earning organizational recognition.

EDUCATION

Bachelor of Engineering (BE / B.Tech) — Mumbai University

2020

CERTIFICATIONS

- Scaler — Data Science & Machine Learning (DSML)
- Power BI Essentials: Power BI Training and Exam Prep

ACHIEVEMENTS

- Promoted to Senior Software Engineer at Loylty Rewardz (Apr 2026) in recognition of consistent performance and delivery.
- Received organizational appreciation for automating the Partner Health reporting pipeline; recognized for end-to-end reporting automation and data-reliability improvements.